6.1 Perpendicular and Angle Bisectors

Name _____ Period ____ Date _____

Geometry 6.1 (30-minute plan) · Student sheet

I can…

- use the Perpendicular Bisector Theorem and its converse to find lengths.
- use the Angle Bisector Theorem and its converse to find lengths and angle measures.
- write an equation of a perpendicular bisector.

Vocabulary · fill in the missing word

Equidistant — the same _____ from two or more objects

Perpendicular bisector — a line, segment, or ray that is perpendicular to a segment at its _____

Distance from a point to a line — the length of the _____ segment from the point to the line

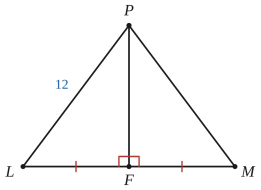
Angle bisector — a ray that divides an angle into two _____ angles

Part 1 · Notes and examples (fill in each note, then work the example)

Perpendicular Bisector Theorem: If a point is on the perpendicular bisector of a segment, then it is _____ from the endpoints of the segment.

Example 1

$\overline{PF}$ is the perpendicular bisector of $\overline{LM}$. Find PM.



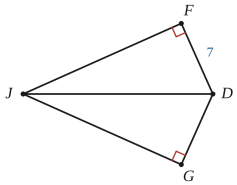
Converse: If a point is equidistant from the endpoints of a segment, then it is on the _____ _____ of the segment.

Angle Bisector Theorem: If a point is on the bisector of an angle, then it is equidistant from the two _____ of the angle.

Distance from a point to a line: The distance from a point to a line is the length of the _____ segment from the point to the line.

Example 2

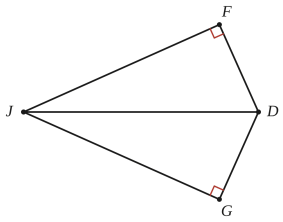
$\overline{JD}$ bisects $\angle FJG$. Find DG.



Converse: If a point is in the interior of an angle and is equidistant from the two sides, then it lies on the _____ of the angle.

Example 3

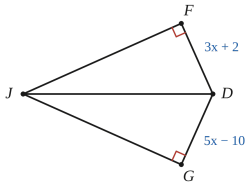
Can you conclude that $\overline{JD}$ bisects $\angle FJG$? Explain.



Part 2 · Partner practice

Partner practice

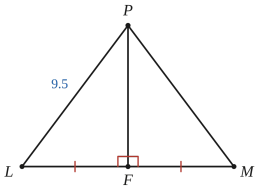
$\overline{JD}$ bisects $\angle FJG$. Find DF.



Part 3 · On your own: choose 2 of 3

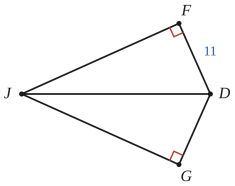
Problem 1

$\overline{PF}$ is the perpendicular bisector of $\overline{LM}$. Find PM.



Problem 2

$\overline{JD}$ bisects $\angle FJG$. Find DG.



Problem 3

Find $m\angle FJG$.

